

File permissions in Linux

Project description

Research team department want to ensure the current research team directory access control is already correct. I will check the research team directory which located at /home/researcher2/projects directory with the correct permission.

Check file and directory details

The first thing to check is where are our current position so the first thing to do is to do pwd, if we not in the directory we will use cd to go there. From the result we was not in the directory of projects so we need to use cd once. To list the directory details we can use ls -l and if we want to check the hidden file we can additionally by changing -l to -la flag.

```
researcher2@86dbed016e95:~$ pwd
/home/researcher2
researcher2@86dbed016e95:~$ ls
projects
researcher2@86dbed016e95:~$ cd projects/
researcher2@86dbed016e95:~/projects$ ls
drafts project_k.txt project_m.txt project_r.txt project_t.txt
researcher2@86dbed016e95:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:32 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 20:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$
```

Describe the permissions string

The first letter from the screenshot above show if the object is a file or a directory, the other 9 letters shown permission for user, group, and other. Which the 1st to 3rd letter was permission for user, 4th - 6th for group, and 6th-9th for other. For the permission itself r is for read, w is for write, and x is for execute.

Change file permissions

The organization does not allow others to have write access to any files. So I will remove other permission for write with `chmod o-w` for a project that still give permission for other to write.

```
researcher2@86dbed016e95:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:32 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 20:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$ chmod o-w project_k.txt
researcher2@86dbed016e95:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:32 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 20:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$
```

From The result the `project_k.txt` still given other permission for write so i remove it with command before.

Change file permissions on a hidden file

The research team has archived `.project_x.txt`, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. From the previous image we can know there is misconfiguration permission for the hidden file so i will fix it by forcing user and group permission to only read by using `chmod u=r,g=r .project_x.txt`.

```
researcher2@86dbed016e95:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@86dbed016e95:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:32 ..
-r--r----- 1 researcher2 research_team  46 Jan  9 20:05 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$
```

Change directory permissions

The files and directories in the projects directory belong to the researcher2 user. Only researcher2 should be allowed to access the drafts directory and its contents. So we need to remove all permission for draft except for the user so we can just remove the group executable permission from the directory with `chmod g-x drafts`.

```
researcher2@86dbed016e95:~/projects$ chmod g-x drafts
researcher2@86dbed016e95:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:05 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 20:32 ..
-r--r----- 1 researcher2 research_team  46 Jan  9 20:05 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jan  9 20:05 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 20:05 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 20:05 project_t.txt
researcher2@86dbed016e95:~/projects$
```

Summary

I already configured all the misconfiguration permission from the projects folder. I use `ls -la` to check the permission and `chmod` to change the permission.